

Darukaa.Earth Knowledge Synthesis Report 07

Deforestation, Habitat Fragmentation and Biodiversity Decline

Document type: institutional scientific synthesis

Topics: deforestation, habitat_fragmentation, biodiversity, land-use change

1. Purpose

Deforestation and fragmentation are among the strongest documented drivers of terrestrial biodiversity loss. This report supports reasoning when users report forest loss, isolated patches, and declining biodiversity.

2. Area, edge and isolation

When forest is cleared, remaining patches become smaller (area effect), more influenced by hot dry edges (edge effect), and more isolated (connectivity effect). Species with large home ranges, strict forest interior needs, or poor dispersal decline first. IPBES (2019) identified land-use change, including deforestation, as a dominant driver of biodiversity decline globally.

Fragmentation is not only about trees. Soil microbial communities, understory plants, and forest-dependent pollinators also change as litter, humidity, and canopy continuity collapse.

3. Cascades to water and climate

Forest loss can reduce local evapotranspiration and alter runoff. Edges are windier and drier, stressing remaining vegetation. This links deforestation to climate stress and water availability even without a regional drought trend.

4. Interventions

- Stop further clearing.
- Restore native vegetation in the shortest gaps between remnants (corridors).
- Prioritise riparian strips, which serve both movement and water-quality functions.
- Protect remnant interior habitat rather than only planting isolated trees in cleared matrices.

Reforestation with a single exotic species can create another monoculture. Native, structurally diverse restoration is the evidence-aligned path for biodiversity.

5. Metrics and time

Habitat diversity and connectivity can begin to improve as corridors establish (years). Species survival for interior specialists may require decades and intact core patches. Do not promise rapid recovery of vanished large vertebrates from corridor planting alone.

6. Source basis

IPBES Global Assessment on land-use drivers; CBD connectivity guidance; FAO FRA context that forest conversion remains a major land-use change process. Pages refer to this synthesis.